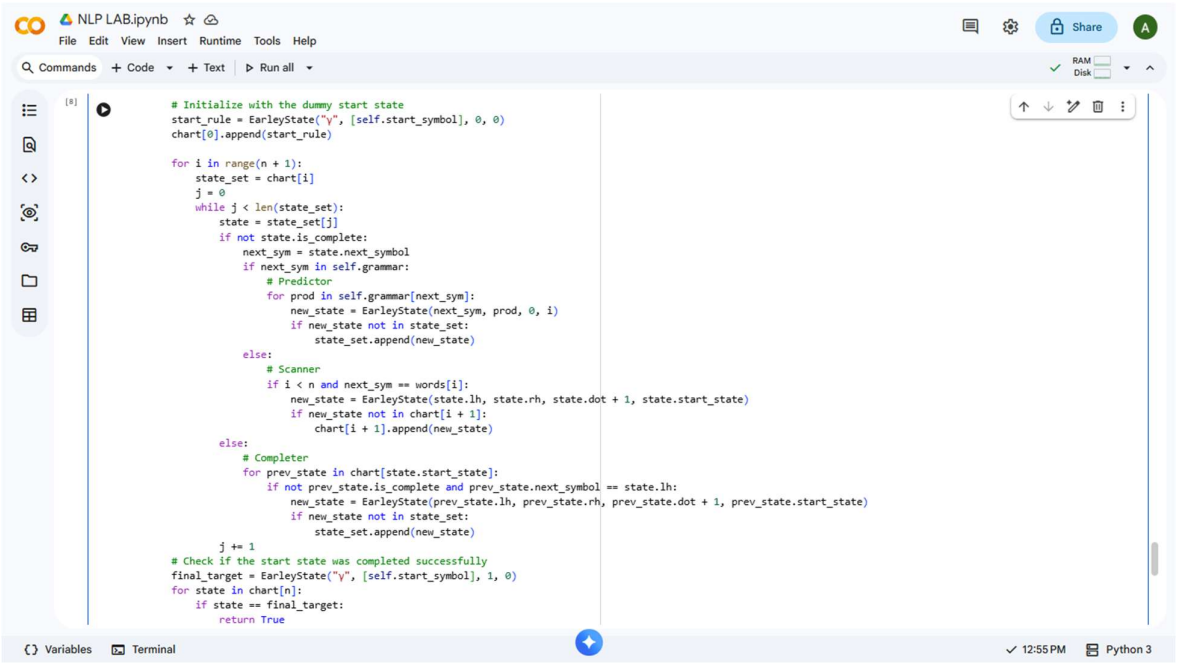
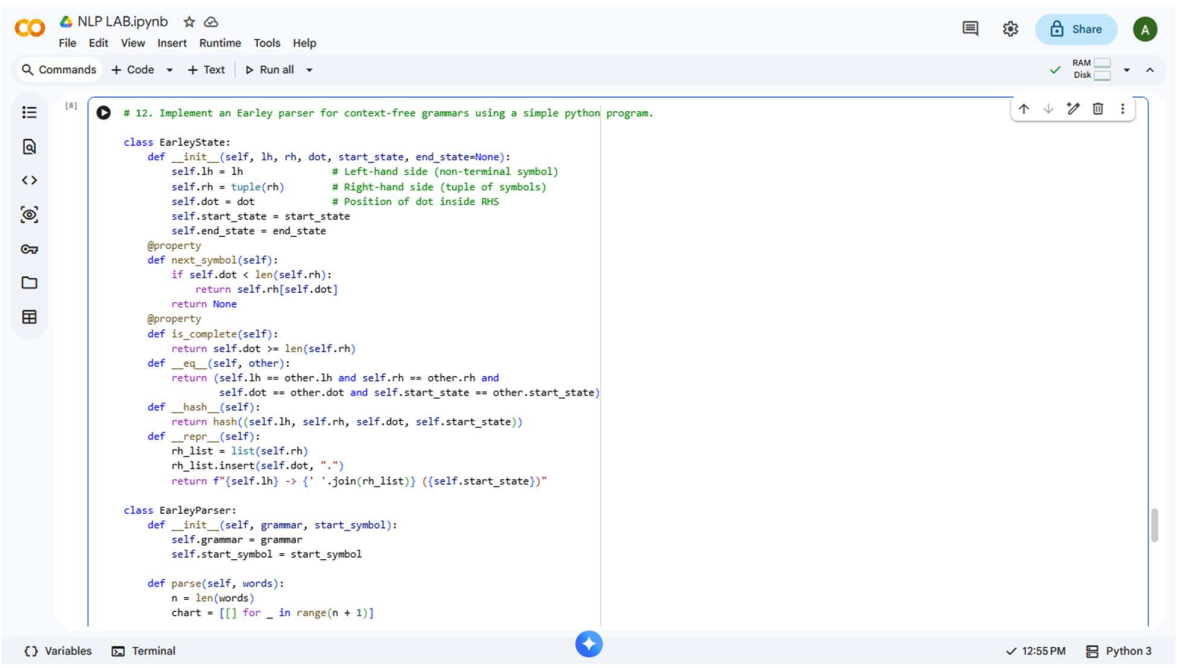
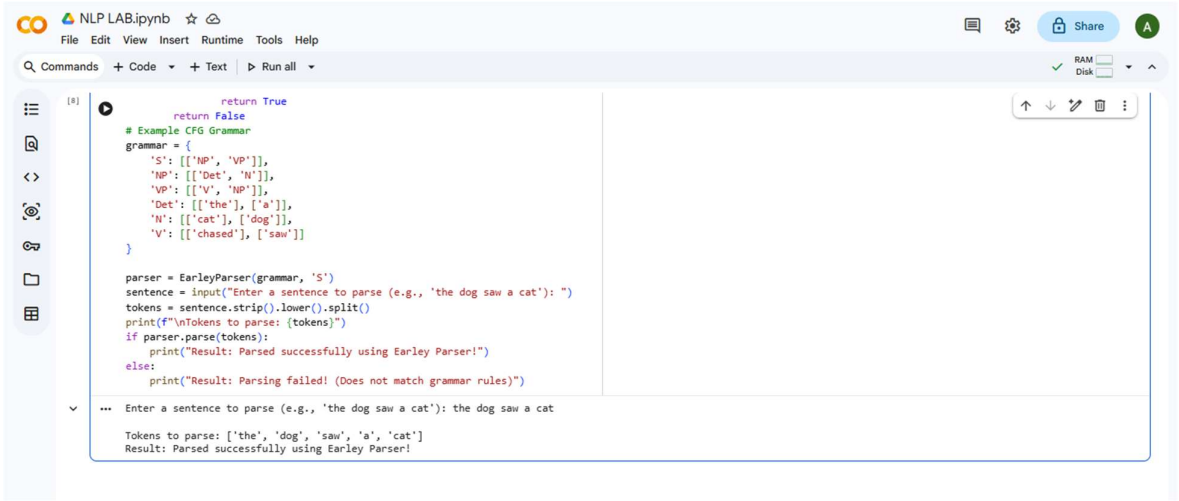
12. Implement an Earley parser for context-free grammars using a simple python program.






NLP LAB.ipynb
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[8]
return True
return False
Example CFG Grammar
grammar = {
'S': [['NP', 'VP']],
'NP': [['Det', 'N']],
'VP': [['V', 'NP']],
'Det': [['the'], ['a']],
'N': [['cat'], ['dog']],
'V': [['chased'], ['saw']]
}
parser = EarleyParser(grammar, 'S')
sentence = input("Enter a sentence to parse (e.g., 'the dog saw a cat'): ")
tokens = sentence.strip().lower().split()
print(f"\nTokens to parse: {tokens}")
if parser.parse(tokens):
print("Result: Parsed successfully using Earley Parser!")
else:
print("Result: Parsing failed! (Does not match grammar rules)")
Enter a sentence to parse (e.g., 'the dog saw a cat'): the dog saw a cat
Tokens to parse: ['the', 'dog', 'saw', 'a', 'cat']
Result: Parsed successfully using Earley Parser!